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1. Summary of Progress

I have completed the baseline evaluation of DeepSeek V4 Pro against the first 20 prompts of the constructed benchmark. Each prompt was submitted independently, with no validation pipeline, and responses were compared against manually established ground truth.

Key finding: DeepSeek V4 Pro performed strongly on factual/lesson-content questions and quiz answer-key questions but exhibited failures on hallucination-trap prompts — most notably Prompt 19 (fabricated concept) and Prompt 20 (fabricated Java API). This confirms that obscurity/plausibility of false premises is a meaningful predictor of hallucination, consistent with the benchmark design assumption.

2. Methodology

- Each prompt was given verbatim to DeepSeek V4 Pro with no additional context.
- Ground truth was established via textbook/curriculum reference (SOLID, GoF patterns, Java standard library docs) or manual verification.
- Responses were scored for: correctness (Correct/Incorrect), hallucination presence (Yes/No), and hallucination type (Lesson content / Obscure terminology / Quiz key / Fabricated concept / Fabricated API).
- No prompt was corrected or clarified mid-response.

3. Baseline Result Spreadsheet

4. Baseline Metrics

Metric	Value
Total prompts	20
Correct responses	18
Hallucinated responses	2
Baseline Hallucination Rate (HR_baseline)	**10.0%**

Metric	Value
False rejection rate (FRR)	N/A (no validation pipeline)

Hallucination breakdown by type:

Hallucination Type	Count
Lesson content	0
Obscure terminology	0
Quiz answer-key	0
Fabricated concept	1
Fabricated API	1

5. Observations

1. **Strong factual grounding:** DeepSeek V4 Pro handled all standard SOLID, GoF, and Java API questions correctly. No hallucinations were observed in Categories A, B (non-fabricated), C, or D.
2. **Vulnerability to false premises:** Both hallucinations occurred in Category E (hallucination-trap questions). The model accepted fabricated premises and generated plausible-sounding but entirely invented details (researcher names, publication years, Java packages).
3. **Obscurity correlation confirmed:** The intentionally obscure but *real* terminology (P6, P7, P8) was handled correctly, suggesting the model's failures are specifically tied to fabricated premises rather than unfamiliar real concepts.
4. **Implication for Confident Agent:** The comparative evaluation should focus on whether the validation pipeline catches P19 and P20 without falsely rejecting correct answers in P1–P18.

6. Next Steps

- Expand benchmark to 50–100 prompts per the recommended distribution (20% basic factual, 30% lesson-content, 20% quiz/answer-key, 15% code reasoning, 15% hallucination-trap).
- Run the identical 20 prompts through the Confident Agent for 8.3 Comparative Evaluation.

- Calculate:
    - HR_agent
    - Hallucination Reduction = (HR_baseline − HR_agent) / HR_baseline × 100
    - False Rejection Rate (FRR) = correct answers incorrectly flagged as untrustworthy / total correct answers × 100
- Record false rejections separately to ensure the Confident Agent is not simply defaulting to "I don't know."